

Hip Arthritis in the Athlete

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Hip arthritis is a common disease. One in four people may develop symptomatic hip osteoarthritis in their lifetime.¹ The treatment of arthritis in the active individual is challenging. The often intuitive recommendation of avoiding pain generating activities and sports seems unthinkable to the athlete. Our understanding of the hip joint has increased substantially in the past 2 decades, with the modern description of femoroacetabular impingement by Ganz and his colleagues and its suspected involvement in the development of osteoarthritis.^{2,3} Corrective procedures for this disorder as well, and others such as acetabular dysplasia, have born the field of hip preservation. Though the efforts of hip preservationists have helped many athletes return to function, and hopefully stave off osteoarthritic changes, solutions for hip osteoarthritis short of arthroplasty remain distant. Hip arthroplasty has continued to evolve, and remains one of the most commonly performed orthopaedic surgeries performed.

HISTORY

Hip arthritis in the young and active patient can often be a challenging entity with regard to selecting appropriate treatment options. Making an accurate diagnosis is the first step in providing suitable treatment options for the patient.^{4,5} Burnett et al. reported that a mean of 3.3 providers were seen prior to a definitive diagnosis being made.⁶ This process can often lead to frustration. A structured history and physical exam can simplify the treating provider's approach to this sometimes complex problem.

Initial patient history should include a detailed description of the patient's current symptoms, including their time of onset, duration, and any inciting trauma or injury. Patients with arthritis often describe a gradual onset of pain, though some may remain asymptomatic until the later stages of disease and experience onset of pain after a particular injury or activity. Aggravating factors such as specific activities or positions should be noted, as should alleviating factors including any already attempted treatments. Patients typically report worsened pain with increased activity such as walking, standing, running, or playing a sport, and improved pain with rest. Sitting with the hip flexed, such as during an extended car or plane ride, or entering or exiting a low vehicle may also aggravate symptoms. The pain often worsens as the day progresses, and can persist as an aching pain at night. The physician's initial task should be to determine whether the patient's pain is indeed intra-articular in origin, as opposed to extra-articular. The location of pain is quite useful

in beginning this differentiation. Intra-articular hip pain is often localized to the deep anterior groin. Patients often use the "C" sign when describing the location of their symptoms.⁷ Though intra-articular pathologies can sometimes present as posterior hip pain, this is much less common. Posterior pain is worthy of considering musculotendinous strains, deep gluteal space syndrome, sacroiliac joint pathology, or lumbar spine pathology. Pain localized to the lateral hip is often the result of greater trochanteric pain syndrome (GTPS). The term GTPS was coined as a less specific but often more precise description for the etiology of lateral hip pain. It encompasses trochanteric bursitis, gluteus medius and minimus tendinopathy or tearing, and snapping iliotibial band or iliotibial band friction syndrome.⁸ Anterior hip pain may also represent musculotendinous injuries, including to the hip flexor musculature, core muscle injuries (including injury to the rectus abdominis insertion and adductor longus origin), or pubic symphysis pathology such as osteitis pubis.⁹ In the athlete, consideration should also be given to pelvic or proximal femoral stress fractures.^{5,10-12}

Patients with hip arthritis often report associated stiffness. Substantial limitations in motion may be a sign of later stages of arthritis. Mechanical symptoms such as catching, clicking, or locking may be experienced, and may represent the presence of loose bodies or chondral injuries, or alternatively snapping of musculotendinous structures.

PHYSICAL EXAM

Examination begins with evaluation of the patient's gait. An antalgic gait is noted in patients with various hip pathologies, including intra-articular issues such as hip arthritis. The antalgic gait is hallmarked by a decrease in the duration of the stance phase of the ipsilateral hip. This effectively decreases the amount of time the patient is applying his weight through the affected hip. Another common gait abnormality is the Trendelenburg gait. This is seen in patients with abductor injuries, including those to the gluteus medius and gluteus minimus tendons. During the stance phase of the affected hip, the contralateral hemipelvis is noted to drop. A compensatory thoracolumbar lurch is noted toward the affected hip to maintain coronal balance.¹³ In patients with hip arthritis, a decreased stride length may also be noted secondary to diminished range of motion (ROM) of the hip. Gait parameters have been shown to correlate highly with radiographic femoroacetabular impingement (FAI) morphology

Abstract

Hip arthritis is a common disease. The treatment of arthritis in the active individual is challenging, and should initially include attempts at conservative measures, include rest, activity modification, and nonsteroidal antiinflammatory medications. Guided injections to the hip joint and periarticular tissues may be helpful to alleviate symptoms. A systematic approach facilitates assessment of the hip, with much information gleaned from the patient history, physical examination, and plain radiographs. Though not always necessary, advanced imaging can be helpful to elucidate early degenerative changes of the hip. Femoroacetabular impingement (FAI) has been described as a precursor to hip osteoarthritis. Its early recognition and appropriate treatment may potentially alter the natural progression of the disease. Use of hip arthroscopy for the treatment of FAI and chondrolabral injuries has demonstrated success, though its utility in the arthritic hip is highly limited. Moderate to advanced hip osteoarthritis is more adeptly treated with arthroplasty.

Keywords

hip osteoarthritis
femoroacetabular impingement
Tönnis grading for hip arthritis
hip chondromalacia
hip arthroscopy
total hip arthroplasty

as well.¹⁴ General observation of the patient may also offer some insight into the underlying cause of pain. The overweight or obese patient may understandably be predisposed to earlier arthritis in the hip due to increases in joint reactive forces, and this factor should be included in the overall assessment and treatment of the patient. The patient's sitting posture often reveals avoidance of deep hip flexion both in cases of nonarthritic intra-articular pain such as femoroacetabular impingement, or cases of hip arthritis.

Examination of the hip then begins with assessment of any skin lesions such as contusions or scars from previous surgical procedures. The log roll test is performed with the patient in the supine position. This simple maneuver consists of rolling the limb gently back and forth between internal and external rotation, avoiding drastic end range motions. This maneuver avoids stressing the musculotendinous structures that surround the hip, instead simply rotating the femoral head within the acetabulum. The test is not very sensitive, though it is specific for pain from intra-articular pathologies. Basic ROM measurements should then be performed. A standardized approach to this is important and leads to reproducible more measurements. Measurements can be performed in the supine position. Flexion, extension, internal rotation, and external rotation are noted. Internal and external rotation measurements are typically made with the hip flexed 90 degrees.^{5,15,16} It is important to ensure that the patient's lumbar spine remains flat on the table, and that the pelvis is not rotating throughout this testing, as this can alter measurements obtained. Comparing ROM measurements to the contralateral side is valuable, as each patient has a unique normal baseline ROM. Patients with intra-articular pathology may show globally decreased ROM. Patients with femoroacetabular impingement tend to have decreased internal rotation.¹⁷ More specifically, individuals with cam morphology of the femoral head-neck junction have more limited ROM than those without cam morphology.¹⁸

Palpation of bony landmarks and soft tissue structures about the hip and pelvis is a simple and useful part of the physical exam to help identify extra-articular sources of pain. A structured approach to this is simple and efficient and plays an important role in guiding treatment. Although both the patient and treating physician's focus often is directed at underlying joint disorders, including arthritis, identification, and improvement of associated soft tissue symptoms can be useful. The adductor longus origin is palpated in the groin with the patient in the supine position with the hip flexed to approximately 45 degrees, and externally rotated. Asking the patient to adduct the thigh against resist, thus contracting the adductor musculature, can increase the sensitivity of this test. Pain may be indicative of adductor longus tendinopathy, or sometimes adductor tendon partial or complete tearing. The pubic symphysis and pubic rami may also be palpated, and specific tenderness here may indicate osteitis pubis. With the patient still in the supine position, the rectus abdominis near its insertion at the superior pubic rami is palpated while the patient is asked to perform a resisted sit up. Tenderness elicited here may represent a core muscle injury. In the setting of hip arthritis, with limited motion of the joint, increased strain may be placed on the rectus abdominis–adductor

aponeurosis as the patients unknowingly compensate for limited joint motion with increased lumbopelvic and abdominal rotation. The proximal anterior thigh hip is palpated to screen for hip flexor strains. The patient is then asked to move to lateral decubitus position. The greater trochanter is palpated. Tenderness here is common and may represent isolated peritrochanteric symptoms or secondary symptoms often seen with intra-articular disorders. The patient's sacroiliac joint and lumbar spine is then palpated. If posterior pain is reported, palpation of the ischial tuberosity and hamstring origin can help narrow the differential diagnosis. Posterior hip pain presents a diagnostic dilemma, though more recent descriptions of deep gluteal syndrome, a nondiscogenic, extrapelvic entrapment of the sciatic nerve, has offered some guidance to the assessment of this group of patients.^{19,20}

Strength testing is then performed. While in the supine position, hip flexion and adduction strength is tested. In the lateral decubitus position with the knee slightly flexed to relax the ilio-tibial band, abduction strength is tested. Muscle strength is graded using the traditional five-point scale. Patients with chronic hip joint pain have been reported to have reduced hip muscle strength in all muscle groups or the ipsilateral hip and to a lesser degree the contralateral hip.²¹

Multiple special tests exist that are helpful in identifying the etiology of pain about the hip. The “Impingement Test,” performed by placing the hip in flexion, adduction, and internal rotation (FADIR), has been used to screen for femoroacetabular impingement. The motion brings the anterosuperior femoral-head junction in close proximity to the anterosuperior acetabular rim. In patients with pincer and/or cam morphology, or those with anterosuperior labral pathology, conflict between the acetabulum and femoral head-neck causes pain. The maneuver is not specific and may cause with a number of intra-articular hip pathologies, including hip arthritis. The contralateral side should also be examined, as an aggressive maneuver may cause some degree of discomfort in even a normal, asymptomatic hip. The FABER (flexion, abduction, external rotation) maneuver is the performed. The ipsilateral ankle may be rested on the contralateral knee while in the supine position, and gentle downward pressure applied to the ipsilateral knee. The maneuver may elicit symptoms in various regions, including the ipsilateral hip joint, ipsilateral posterolateral hip soft tissues, and the contralateral sacroiliac joint. Ipsilateral hip joint pathology typically causes deep anterior pain with this maneuver. It is important to note location of pain, as opposed to simply its presence. The posterior impingement test is performed by abducting, extending, and externally rotating the hip. Unlike the FADIR maneuver, it brings the posterolateral femoral head-neck junction in conflict with the posterior acetabular rim, potentially causing pain. It may also cause pain in a hip with limited ROM, or one with adhesive capsulitis. Patients with arthritis often have globally diminished ROM, and this maneuver may be painful.^{5,13,16}

The Stinchfield test, a resisted straight leg raise performed at approximately 30 degrees of hip flexion with the patient supine, may cause deep anterior hip pain in patients with intra-articular pathology. However, this is not a specific test, as those with hip

flexor strains or other musculotendinous injury will likely also have pain with it.

In summary, physical examination of the hip with arthritis can be expected to show globally diminished ROM; the patient may commonly have sites of tenderness to the extra-articular musculotendinous structures of the hip and pelvis; the log roll, FADIR, FABER, Stinchfield, and posterior impingement tests may elicit pain. The patient may present with an antalgic gait pattern.

IMAGING

Imaging of the hip begins with plain radiographs. In cases where the clinical presentation and plain radiographs fit the diagnosis of arthritis, further imaging beyond plain radiographs is often unnecessary. Plain radiographs of the hip should begin with a well centered anteroposterior (AP) view of the pelvis. This is most commonly obtained in the supine position, but may also be obtained in the standing position to provide information about the patient's functional sagittal pelvic tilt.^{22,23} The coccyx and pubic symphysis should be in line, with the caudal tip of the coccyx between 2 and 3 cm from the pubic symphysis. The obturator foramen should be symmetric bilaterally (Fig. 88.1).²⁴ Variations in pelvic tilt, thus providing oblique views with characteristics of the inlet or outlet view, have been shown to skew interpretations of acetabular version. When the pelvis is tilted anteriorly, the acetabulum may appear falsely retroverted, and when the pelvis is tilted posteriorly, the acetabulum may appear falsely anteverted. Rotation of the pelvis can similarly skew measurements of acetabular version as well as acetabular depth. On the anterosuperior measurement, general observation of any joint space narrowing should be assessed. The joint space should be measured, as the severity of joint narrowing has been inversely correlated with success after treatment with hip arthroscopy. Joint space of less than 2 mm has been reported as a lower limit, below which arthroscopy should be avoided for treatment. Measurements are routinely made in three locations: superomedially,

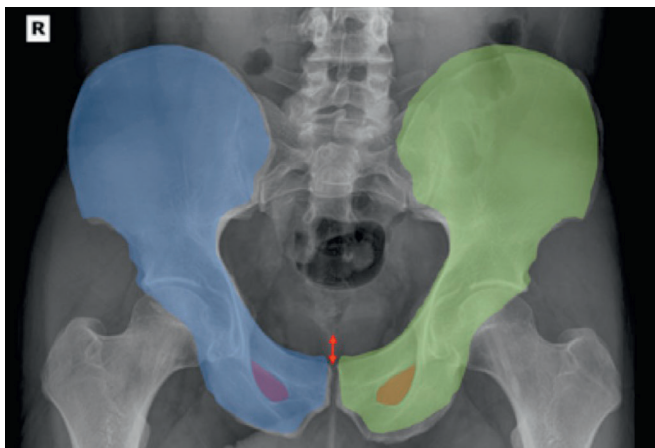


Fig. 88.1 Anteroposterior view of the pelvis. A well-centered view is important, including symmetry of the bilateral hemipelvis, obturator foramen, and a distance of 2 to 3 cm between the coccyx and pubic symphysis.

superiorly, and superolaterally. The presence of subchondral sclerosis of the acetabulum or femoral head should be noted, as should the presence of osteophytes. Subchondral cyst formation in the acetabulum or femoral head, when visible on plain radiographs, typically indicates more advanced degenerative changes. Fig. 88.2 demonstrates a hip with these advanced arthritic findings. Degenerative changes are often classified using the Tönnis grading system (Table 88.1).²⁵ The Lateral Center Edge Angle of Wiberg is then measured (Fig. 88.3). This provides a measurement of lateral acetabular coverage of the femoral head. A best fit circle of the femoral head is drawn, and an angle from its center with one limb directed superiorly and the other to the lateral margin of the acetabulum is drawn. Conventionally, acetabular dysplasia has been diagnosed when the lateral center edge angle is less than 25 degrees. More recently, a subcategory of “borderline” dysplasia has been described in patients with LCE measurements of 18 to 25 degrees. Acetabular dysplasia has important implications for treatment of hip pain, as patients with true dysplasia are typically not good candidates for hip arthroscopy, and may indeed require pelvic osteotomies to correct



Fig. 88.2 An anteroposterior view of the left hip showing severe joint space narrowing, femoral and acetabular sclerosis, and subchondral cyst formation. These changes are consistent with Tönnis grade 3.

TABLE 88.1 Tönnis Classification of Osteoarthritis

Tönnis Grade	Imaging Characteristics
0	No signs of osteoarthritis
1	Slight narrowing of the joint space, slight osteophyte formation, and slight sclerosis of the femoral head or acetabulum
2	Small cysts in the femoral head or acetabulum, increased narrowing of the joint space, and moderate loss of sphericity of the femoral head
3	Large cysts, severe narrowing or obliteration of the joint space, severe deformity of the femoral head



Fig. 88.3 The lateral center edge angle is used to measure lateral acetabular coverage of the femoral head.

their underlying bony acetabular deficiency if substantial arthritis is not present.

The presence of an acetabular crossover sign is evaluated on the AP plain radiograph. The crossover sign is present when the anterior acetabular wall and posterior acetabular wall cross prior to their convergence at the lateral acetabular sourcil. This indicates some degree of retroversion of the acetabulum, sometimes a contributing factor in femoroacetabular impingement. The further distal/medial the point of crossover, the more substantial the retroversion.²⁶ The crossover sign can be quantified as a ratio of the distance between the lateral margin of acetabular sourcil and the crossover point to the diameter of the acetabulum altogether. The posterior wall sign is present when a point marked at the center of the femoral head lies lateral to the posterior wall of the acetabulum on the AP radiograph, and may indicate undercoverage of the posterior acetabulum, which can exist as part of a retroverted acetabulum or in the setting of global acetabular undercoverage.

Various lateral views of the hip exist. Commonly used examples include the frog-leg lateral, the cross-table lateral, and the Dunn 45-degree or Dunn 90-degree views. Each view may offer a slightly different angle and view of the femoral head-neck junction. The frog-leg lateral is a technically simple and easily reproducible. In many hip arthroscopy and preservation clinics, the Dunn 45-degree lateral view is utilized, given that it has been shown to be most sensitive for the presence of CAM morphology.²⁷ Regardless of which view is utilized, the presence or absence of CAM morphology should be noted, and when present it should be quantified. The CAM morphology can be quantified using the alpha angle. The alpha angle was classically described using axial magnetic resonance imaging (MRI) cuts, but can be extrapolated to lateral radiographic views.²⁸ This is measured by first drawing a best fit circle on the femoral head. At the center of this circle, an angle with one limb directed parallel to the femoral neck and the other directed to the point where the femoral head-neck exits the best fit circle (Fig. 88.4). Gosvig et al. classified alpha angle measurements as pathologic (>57 degrees), borderline (51 to 56 degrees), subtle (46 to 50 degrees), very subtle (43 to 45 degrees), or normal (≤ 42 degrees).²⁹ Commonly alpha angle

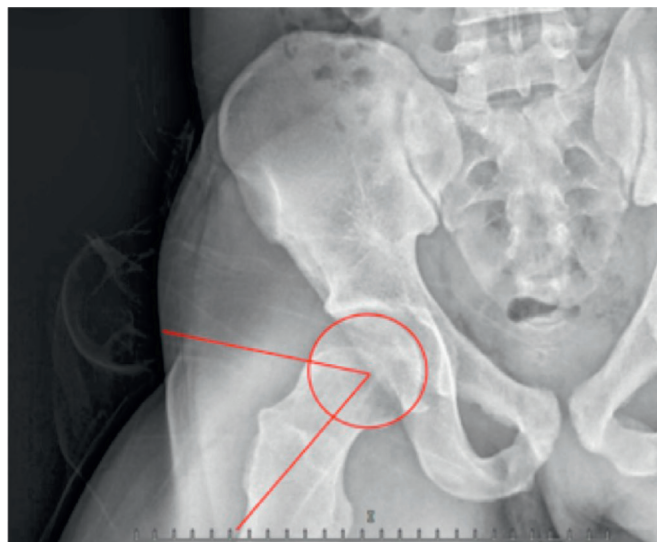


Fig. 88.4 The alpha angle is used to quantify the cam morphology at the femoral head-neck junction.

values greater than 50 degrees are generally regarded as abnormal, and represent CAM morphology.³⁰ Another available method of quantifying CAM morphology is the femoral offset. Again using the center of the femoral head on the lateral view, a line is drawn in the axis of the femoral neck. Two other parallel lines are drawn—one at the femoral neck and the other at the anterior-most border of the femoral head. A decreased offset is seen in patients with cam morphology. As an alternative to reporting the true offset, the offset ratio may be obtained by dividing the offset by the diameter of the femoral head, and may offer a more standardized value.²⁴

MRI is commonly used to aid in the evaluation of the athlete with hip pain. It can be obtained with or without intra-articular contrast (arthrogram). Magnetic resonance arthrogram (MRA) has excellent sensitivity in identifying labral tears at the chondrolabral transition zone. Debate exists on the necessity of arthrogram for assessment of chondrolabral structures of the hip.^{31,32} Improvements in MRI technology and the use of 3.0 T scanner have led to many hip preservation surgeons avoiding the routine use of the arthrogram. 3.0 T nonarthrogram MRI has shown excellent sensitivity and specificity in identifying arthroscopically proven labral tears.^{33,34} In addition to assessment of the labrum, MRI is a sensitive imaging modality able to identify early degenerative changes, including subchondral edema and subchondral cyst formation. Acetabular and femoral head chondral irregularities, thinning, or delamination can be evaluated. The clinical relevance of this often depends on the degree of arthritis. One must take care to avoid focusing on the presence or absence of labral pathology. Subchondral edema may indicate early stages of degenerative disease, and may be masked by the stark contrast created by intra-articular contrast, thus offering an additional reason to consider nonarthrogram studies. In a study of 208 hip patients over 50 years of age, MR arthrogram identified some degree of labral pathology in 93% of patients and true labral tearing in 73% of patients (with no statistical correlation between labral tearing and presence/degree of arthritis).³⁵

Patients with diagnosed labral tears may be quite determined that repair of this structure will rid them of their symptoms. Treatment of a torn acetabular labrum in the arthritic hip is unlikely to provide substantial mid to long-term improvement, and should be avoided. The ligamentum teres is also visualized, with surrounding synovium in the cotyloid fossa. Damage to the ligamentum teres or synovitis in the cotyloid fossa and diffusely throughout the joint may be seen. Loose osteochondral bodies may also be present in the setting of arthritis, and may be responsible for some mechanical symptoms.

Evaluation of the periarticular soft tissues may also yield important information about other extra-articular sources of pain. Correlation to the physical examination is important, as sites of musculotendinous edema, or even low grade partial thickness tearing, may not always be clinically relevant, and are commonly noted around the degenerative joint. Pathology of the gluteus medius or gluteus minimus tendons at the respective insertions at the greater trochanter should be evaluated, along with adjacent greater trochanteric bursitis. Tears of these abductor tendons are clinically significant when the patient's pain can be localized to the lateral hip at the greater trochanter, and in these cases should be treated. Peritrochanteric pathology and intra-articular pathology may be compensatory to each other; thus thoughtful assessment and treatment of the underlying cause of pain as well as associated symptoms may yield the best results. Femoral neck and acetabular stress fractures should be ruled out on MRI, and are especially common among longer distance runners and other impact activities. These may not be visible on plain radiographs but easily noted on MRI. They may be present concurrently in a runner with arthritis, but may cause acute chronic pain in the athlete with known arthritis of the hip.

Computed tomography (CT) is not usually necessary for diagnostic purposes around the hip in the setting of FAI or osteoarthritis. It does have a role for preoperative planning in patients who have an established diagnosis and are indicated for surgery. Two-dimensional radiographs also do have some limitations, and the use of the measurements such as lateral center edge angle and anterior center edge angle may not tell the whole story as it relates to acetabular coverage. Similarly, femoral head-neck convexities and irregularities exist in variations that two-dimensional radiographs do not fully represent. Three-dimensional reconstruction of CT imaging is now possible, providing easily manipulated images that morphology aberrances such as CAM or pincer morphology, thus guiding surgical treatment. Various commercially available software options exist that allow motion assessments of the hip, indicating areas of bony conflict between the acetabulum and femoral head in FAI.

DECISION-MAKING PRINCIPLES

In keeping with the typical goals of treating the active patient, the treatment of athletic patients with hip arthritis aims at minimizing pain or discomfort while keeping the patient active. Unlike many other conditions diagnosed and treated by sports medicine specialists, arthritis carries certain weight and implications for the patient. Many patients view it as an end to their ability to either compete in their sport or enjoy it recreationally. In general,

regardless of the severity of arthritis on imaging, treatments should be begun with noninvasive to minimally invasive measures, directed at reducing pain and limiting the progression of degenerative changes.

TREATMENT OPTIONS

Nonsurgical Measures

Nonsurgical treatment of hip arthritis includes the use of medications to relieve pain and physical therapy to optimize ROM and strength of the hip and pelvic musculature. Nonnarcotic analgesics such as acetaminophen or nonsteroidal antiinflammatory drugs (NSAIDs) can provide relief. Both categories of medications should be prescribed and taken with knowledge of dosing guidelines. Current guidelines recommend against the use of greater than 3000 mg of Tylenol per day to avoid hepatic injury. NSAIDs may need to be taken with gastroprotective agents, especially in patients with a history of gastroesophageal reflux, gastric ulcers, or upper gastrointestinal bleeding. Narcotics are avoided due to their potential for the development of dependency or addiction. The use of narcotics also is contraindicated while driving and operating machinery. Nationwide efforts to reduce the misuse of narcotics have also been launched, and an effort should be made to avoid prescribing them unless absolutely necessary. In cases of acute injury or pain, a period of relative rest may be useful. In recreational athletes, activity modification, including temporary or long-term transition from high impact activities such as running and jumping to lower impact activities such as cycling, elliptical treadmill, and swimming, should be attempted. Manual therapy with myofascial mobilization may be beneficial to improve soft tissue symptoms about the hip. Cryotherapy and/or heat therapy may also provide relief of soft tissue symptoms.

Injections may provide a useful adjunct to treatment, both from a diagnostic and therapeutic standpoint. The use of the ultrasound in musculoskeletal medicine has allowed for precise placement of injections, thus allowing clinicians to more reliably target areas or structures responsible for pain. Common injection sites include the hip joint itself, the iliopsoas bursa, the peritrochanteric bursa, the hamstring origin, the piriformis muscle/tendon, and the sacroiliac joint. In the patient with hip arthritis, intra-articular injections are useful to assess how much of the patient's pain is directly related to the degenerating joint versus the result of compensatory soft tissue dysfunction about the hip and pelvis. Injections are typically performed with a local anesthetic and a corticosteroid. Whereas the diagnostic utility of the local anesthetic is likely more relevant in the non-arthritic hip, the therapeutic benefits of the corticosteroid may be of the utmost importance in the arthritic joint. Ultrasound guided diagnostic injections tend to be less painful and more convenient for patients than those performed under fluoroscopic guidance.³⁶ Viscosupplementation has been utilized in the hip with some evidence of improved symptoms in hip osteoarthritis.^{37,38} Though pain may be improved in the short term, the natural course of hip osteoarthritis has not been shown to be altered by viscosupplementation. More research will be necessary prior to broadly recommending such treatment.

SURGICAL TREATMENT

Hip Arthroscopy

The role of arthroscopy in treating disorders of the hip has greatly increased in recent years.^{39,40} This increase is largely attributable to the understanding and more frequent treatment of femoroacetabular impingement. Despite this, arthroscopy has a limited role in the treatment of arthritis once it is present. Arthroscopy should not be viewed as a tool to treat hip arthritis. Short-term and now long-term studies have shown that patients with arthritis have poorer results after hip arthroscopy.⁴¹⁻⁴⁶ A study of patients undergoing hip arthroscopy including 43 patients with Tönnis grade 2 changes showed that those with grade 2 Tönnis changes had a 7.73 odds ratio of undergoing total hip arthroplasty (THA; compared to those with Tönnis grade 0 or 1 changes) at minimum follow-up of 2 years.⁴⁷ The role of arthroscopy lies in the treatment of irregularities thought to represent precursors to arthritis. Femoroacetabular impingement has been described as a predisposing lesion to hip osteoarthritis. The presence of CAM morphology leads to delamination of adjacent acetabular cartilage, thus beginning the cascade of arthritic changes. Pincer morphology may have a more important role as a precursor to labral pathology. The labrum has an important role as a regulator of fluid dynamics within the hip joint; when torn or resected, the negative pressure seal of the hip is compromised and increased joint reaction forces are noted, potentially leading to arthritis.⁴⁸ Fig. 88.5 shows an anterosuperior hip joint with disruption at the chondrolabral junction. Early adjacent chondromalacia is noted adjacent to the labral tear. Hip arthroscopy offers the ability to correct the bony irregularities in FAI, as well as stabilize the labrum and chondrolabral junction by repairing the labrum and resecting any unstable chondral flaps. Microfracture of the acetabulum chondral lesions in association with the treatment of FAI when necessary has also yielded promising results.⁴⁹⁻⁵¹ Though labral repair has been regarded as a superior option over labral débridement due to improved functional outcome scores at short term follow-up, focal/selective débridement remains a reasonable option when



Fig. 88.5 Arthroscopic view of the anterosuperior chondrolabral junction in a patient with femoroacetabular impingement. There is early chondral fraying and softening adjacent to the area of labral tear.

labral tissue is too friable and degenerated to allow for repair. Complete excision of the labrum to bone is not recommended, as this has shown worse patient reported outcomes.⁵² Comparison of labral repair versus débridement has actually shown fairly similar outcomes at 10-year follow-up, with a slight advantage to labral repair. Capsular management has gained interest. Repair of capsulotomies has been associated with improved patient reported outcomes after hip arthroscopy, lending credence to the important role of fluid dynamics in the hip joint, and the important role of the thick capsular ligaments of the hip in preventing microinstability of the hip.^{53,54} In patients with borderline acetabular dysplasia, capsular plication associated with labral preservation (repair) has been successful in the short term.⁵⁵ Longer term evaluation of the patient's hip will be important. In general, hip arthroscopy should be either avoided or entertained with extreme caution in patients with degenerative changes beyond Tönnis grade 1. Those patients with some degenerative changes that are treated with arthroscopy should be educated that their postoperative improvements may be less substantial, and that the rate of conversion to hip arthroplasty is higher. Endoscopic techniques have also evolved and now also allow the treatment of extra-articular disorders such as gluteus medius and minimus tears, proximal hamstring tendon pathology, and posterior deep gluteal space pathology.

Osteotomy

Hip arthroscopy should be avoided in patients with overt acetabular dysplasia. Whereas arthroscopic techniques allow surgeons to correct pincer and CAM morphology by removing excessive bone, bone cannot be added in cases of insufficient acetabular coverage. The Bernese periacetabular osteotomy (PAO) is utilized to treat acetabular dysplasia. This procedure is performed by making a single incision and several bony cuts in the innominate bone prior rotating the acetabulum to provide greater coverage of the femoral head and then fixed in its new position using screws.^{56,57} As with arthroscopy, the technique is most successful in the absence of advanced arthritic changes. The procedure is typically performed by surgeons specializing in joint preservation, and is not without its complications. In the presence of chondrolabral injury or cam morphology of the femoral head-neck junction associated with acetabular dysplasia, arthroscopy for labral repair and femoroplasty has been described in the same setting as PAO, with good results.⁵⁸ Proximal femoral dysplasia requiring osteotomy represents a much smaller subset of the pathology encountered, but in certain cases may be beneficial.

Hip Resurfacing

Historically hip resurfacing was a preferred surgical option for young patients who wanted to remain active. The advantage to hip resurfacing is preservation of proximal femoral bone stock in the younger patient population to make subsequent revision less invasive. In 2006, the Birmingham Hip Resurfacing was approved by the FDA for use in the United States; however, use has recently fallen out of favor due to concerns associated with metal on metal bearing surfaces and failure rates. Moreover, recent studies have demonstrated no functional benefit to the Birmingham Hip Resurfacing over traditional THA.⁵⁹

Total Hip Arthroplasty

In a young and active patient, joint preservation remains the goal. However, when degenerative changes in the hip preclude preservation of the joint, THA is the gold standard treatment. THA is reserved for patients with end-stage degenerative changes of the hip joint (e.g., bone on bone articulation) that have failed all conservative measures, including nonsteroidal anti-inflammatories, formal physical therapy, intra-articular corticosteroid injections, and have pain affecting their daily activity. Previously THA was reserved for older and more sedentary patients; however, recent improvements in materials and implant fixation have expanded the indication for THA to younger patients. Studies have demonstrated that delaying THA in an appropriately indicated patient actually diminishes the chance of obtaining a satisfactory result.⁶⁰

The most commonly used approaches for THA include the posterolateral approach, anterolateral approach, and direct anterior approach. The optimal surgical approach is at the discretion of the operating surgeon and should be tailored to each individual patient, taking into account factors such as retained hardware, abnormal anatomy, body mass index, and flexion contractures. There is evidence that the direct anterior approach has expedited recovery, including less time using assistive devices after surgery and lower dislocation rates even without hip precautions.⁶¹

Cementless fixation in THA is the gold standard in the United States, particularly in younger patients. The benefit of biologic fixation between the implant and bone provides a longer durability of the prosthesis and allows for bone remodeling. Excellent long-term outcomes have been demonstrated with these components. Selection of the bearing surface for young active patients undergoing THA is an important consideration. Factors to consider include wear characteristics, maximum head diameter, and potential complications from wear over time. Currently the bearing of choice for young patients for most arthroplasty surgeons is a ceramic head with a modern generation highly cross-linked polyethylene liner. Depending on the cost difference between a metal and ceramic head at any given institution, the use of a ceramic head becomes cost effective typically around the age of 65 years or younger.⁶² Metal on metal THA has largely fallen out of favor for concerns with pseudotumor formation and metallosis. Ceramic on ceramic bearings have excellent wear characteristics; however, there have been cases of liner fracture as well as squeaking.

POSTOPERATIVE MANAGEMENT

Hip Arthroscopy

Rehabilitation after hip arthroscopy is widely variable depending on surgeon preference and physiotherapist experience. Most agree that early ROM, including circumduction of the hip, can help avoid stiffness and postoperative adhesions.⁶³ Structured rehabilitation protocols are typically broadly divided into four phases: (1) mobility and initial exercise, (2) intermediate exercise and stabilization, (3) advanced exercise and neuromotor control, and (4) return to activity.⁶⁴⁻⁶⁷ Patient progression through these phases is highly variable, and most recommend progression based on

individual symptoms and abilities. Return to running can be undertaken through a running progression program, with patients monitoring and categorizing their symptoms, and determining progression based on these symptoms. Functional abilities and the presence and degree of pain seem to be better regulators of the patient's ability to return to running than time from surgery.⁶⁸ In a survey of high-volume hip arthroscopy centers, 70% of surgeons recommended waiting 12 to 20 weeks before returning to sport. A total of 85% of surgeons recommended criteria of pain free running, jumping, lateral agility drills, and single-leg squats be used prior to returning to sport.⁶⁹

Total Hip Arthroplasty

In general, the postoperative course after THA allows the patient to be full weight bearing immediately after surgery. There may be motion restrictions based on the approach; however, the direct anterior approach to THA results in no restriction. Patients typically graduate from ambulation with a walker to a cane as soon as they feel comfortable and then to no assistive device. Most surgeons limit high-impact activity, including heavy lifting, jogging, or running, for a period of time after the operation to allow for sufficient osseointegration of the implants. Many surgeons discourage running or jogging after THA to decrease the risk of early failure. This can be challenging, particularly in the young patient population; however, modern implants may be able to better withstand higher impact activity.

RESULTS

Several studies now exist showing very short-term outcomes after hip arthroscopy for treatment of labral tears and femoroacetabular impingement. Consider the 2009 systematic review by Clohisy et al., with 11 studies (nine level 4 and two level 3) with a mean follow-up of 3.2 years. Reduced pain and function was noted in all studies (range 68% to 96%). The study did note the need for longer term follow-up to determine survivorship of the procedure and the impact it could have on the natural progression toward osteoarthritis.⁷⁰ Outcomes of hip arthroscopy, specifically in athletes, have been examined and are shown to be excellent. Nho et al. revealed significant improvement in Modified Harris Hip Scores and Hip Outcome Scores in their series of 33 elite athletes and that 79% of patients had returned to play sports at 1 year, with 73% playing sports at 2 years postoperatively.⁷¹ Philippon et al. published outcomes on hip arthroscopy for labral repair and treatment of FAI in professional hockey players, showing high satisfaction at 2 years postoperatively and improvement in the mean mHHS from 70 to 95.⁷² Studies with longer term follow-up are fewer in number, but share present information. Byrd et al. reported the outcome of 26 patients treated for acetabular labral tears with 10-year follow-up. They reported a mean improvement in the mHHS of 29 points (from 52 preoperatively to 81 postoperatively). Perhaps the most valuable lesson is noted when the cohort was further divided into patients with and without arthritis. Among the 18 patients without arthritis, 15 (83%) continued to have substantial improvement at 10-year follow-up. Conversely, 7/8 (88%) of patients with preexisting arthritis were converted to THA at a mean of 63 months.⁴² In

a wider study of 52 patients undergoing hip arthroscopy for a wider range of diagnoses (labral pathology, chondral damage, synovitis, loose bodies), including those of the previous study, the authors again note that the presence of hip arthritis at the time of the index procedure was a poor prognostic factor.⁷³ Skendzel et al. specifically examined preoperative hip joint space on plain radiographs as a predictor of outcomes. Their study examined 466 patients at a minimum of 5 years follow-up; 54/63 (86%) patients with joint space less than or equal to 2 mm were converted to THA, while only 63/403 (16%) were converted to THA.⁴⁵ Philippon et al. assessed hip arthroscopy for FAI in patients 50 years old or older. Patients with joint space greater than 2 mm fared well, while those with joint space narrowing had increased conversion rate to THA.⁴⁶ Menge et al. reported on a series of 154 patients with 10-year follow-up to compare the outcomes of labral débridement versus labral repair. Both groups showed significant improvements in outcome scores, though joint space less than or equal to 2 mm, increased age (>50 years), and acetabular chondral injury requiring microfracture were independently associated with increased hazard rate for THA.⁴¹

The midterm results of modern day THA are typically reported to be greater than 95% at 5 to 10 years, even in young patients.⁷⁴ However, in young patients, survivorship at follow-up of 13 to 15 years has been reported to drop as low as 54% with earlier generation implants.^{75,76} Further studies with modern generation implants are needed to understand their long-term survivorship.

COMPLICATIONS

Hip arthroscopy is a challenging procedure. Several known complications exist, including traction related, neurapraxias, pudendal nerve neuralgia from compression of the perineal postoperative infection, retroperitoneal fluid extravasation, heterotopic ossification of the periarticular soft tissues, iatrogenic chondrolabral injury, bursitis, deep venous thrombosis, and postoperative femoral fracture. Conversion to hip arthroplasty is also commonly listed as a potential complication. Though in some instances this may be the case, in most this may be the natural progression of existing disease, and thus the importance of patient selection is highlighted. Using a national database study, Truntzer et al. reported the rate of conversion to THA within 1 year at 2.85%.⁷⁷ In a recent 10-year survivorship study of 154 hips by Menge et al., 34% of patients underwent THA.⁴¹ Though age might contribute, most agree that joint space narrowing and the preexistence of arthritis at the time of arthroscopy are most predictive of conversion to THA.

Complications of THA in young patients are similar to those in the older patient population. Common causes of revision are bearing surface wear and aseptic loosening of the acetabular component. In patients with ceramic on ceramic articulation, component chipping or fracture is possible but rare. In a study of 102 THAs performed in patients with a mean age of 20 years, there was a 9% rate of complications resulting in a return to the operating room.⁷⁸ These complications included dislocation, periprosthetic fracture, peroneal nerve palsy, and arterial injury.



Authors' Preferred Technique

In the young patient with intra-articular hip pain that has been and some degree of arthritis, initial treatment comprises initial conservative measures including relative rest/activity modification, physical therapy or a home stretching program, and NSAIDs. Intra-articular injections to the hip can be utilized. If they provide extended relief, these are a reasonable option for the patient who is too arthritic for hip arthroscopy, and younger than preferable for THA. Our preference is to avoid hip arthroscopy in patients with less than or equal to 2 mm of joint space, or with osteophytes secondary to arthritis. We also avoid hip arthroscopy in patients that have notable subchondral edema or cyst formation. We prefer hip arthroscopy for patients with symptomatic FAI and/or labral pathology, without any substantial arthritic changes. We prefer labral repair over débridement if the labral tissue quality allows for this, though partial labral débridement remains a good option in patients with focal labral injuries and friable tissue not amenable to repair. Femoroplasty and acetabuloplasty are performed as needed to correct impingement morphology. Our preferred technique for capsular management involves utilizing an interportal capsulotomy connecting the anterolateral portal and anterior portals, which is repaired at the conclusion of the procedure. For young patients requiring THA, we prefer the direct anterior approach. The patient is placed supine on an operating table that allows manipulation of the extremity during the surgery. The interval between the sartorius and tensor fascia lata is developed. The rectus femoris is retracted medial, and no muscles or tendons are incised. A "T"-capsulotomy is performed and later repaired. Cementless implants are used on both the acetabular and femoral side. Screws are used if necessary through the acetabular component, and a highly cross-linked polyethylene is placed. A tapered design femoral stem is used that is proximally porous coated. A ceramic head is used, which is typically either a 32- or 36-mm diameter. Smaller femoral heads are favorable to allow for thicker polyethylene liner dimensions. Postoperatively the patients have no restrictions on weight bearing or ROM.

FUTURE DIRECTIONS

Hip arthritis is a difficult diagnosis to treat in the active adult. Recent innovations have helped us understand the link between femoroacetabular impingement and hip osteoarthritis, and have given us an opportunity to hopefully change the natural course of this disease. Unfortunately, once the hip has crossed into the realm of arthritis, options for hip preservation quickly narrow. History has shown us in the knee and shoulder that arthroscopy is not generally helpful in the arthritic joint, and this seems to hold true in the hip. Perhaps future developments in biologics will give us more options for restoring articular cartilage. As much as we have learned about femoroacetabular impingement and the possibilities hip arthroscopy offers, it may be that we are still just scratching the surface in this field that is still in its infancy. Arthroplasty generally offers reliable results for the arthritic hip, but certainly presents its own challenges in the younger, more active patient. Although the volume of young patients undergoing THA continues to increase, the long-term results of modern generation implants remain to be seen. The combination of earlier intervention of hip arthritis with THA and longer life expectancy will likely result in an increase in revision THA in the future. Careful surveillance of these patients with routine follow-up is necessary for early intervention.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

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